

SPECIAL QUSOFT SEMINAR · CWI AMSTERDAM

Large Language Models

The industrialization of theoretical science

Tobias J. Osborne · Leibniz Universität Hannover

4 September 2026 · Turing Room, Amsterdam Science Park

PART ONE

Why this talk?

It started as a drop

February 13, 2026 Research Publication

GPT-5.2 derives a new result in theoretical physics

In a new preprint, GPT-5.2 proposed a formula for a gluon amplitude later proved by an internal OpenAI model and verified by the authors.

[Read the preprint ↗](#)



Listen to article | 3:54

[Share](#)

We've published a new preprint showing that a type of particle interaction many physicists expected would not occur can in fact arise under specific conditions. The work focuses on gluons, the particles that carry the strong nuclear force. The [preprint](#) is available on arXiv and is being submitted for publication. In the meantime, we welcome feedback from the community.

arXiv:2602.12176 · Harvard, Vanderbilt, Cambridge, OpenAI · February 2026
gauge-theory scattering amplitudes: GPT-5.2 conjectured the closed form and proved it

[Submitted on 20 May 2026]

Remarks on the disproof of the unit distance conjecture

Noga Alon, Thomas F. Bloom, W. T. Gowers, Daniel Litt, Will Sawin, Arul Shankar, Jacob Tsimerman, Victor Wang, Melanie Matchett Wood

We present a short, digested, human-verified version of the recent OpenAI-generated counterexample to the Erdős unit distance conjecture, and a sequence of reflections on it. The argument relies crucially on ideas that may, at least in retrospect, be attributed to Ellenberg-Venkatesh, Golod-Shafarevich, and Hajir-Maire-Ramakrishna.

Comments: 19 pages; expository note based on very recent developments

Subjects: **Combinatorics (math.CO)**; Number Theory (math.NT)

Cite as: [arXiv:2605.20695](#) [[math.CO](#)]

(or [arXiv:2605.20695v1](#) [[math.CO](#)] for this version)

<https://doi.org/10.48550/arXiv.2605.20695> 

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References & Citations

[NASA ADS](#)

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The trickle is becoming a flood

T1

machine-checked or officially graded

T2

arXiv: named humans read it

T2+

refereed journal

T3

announced only, no referee

T1

Jan 2026

Erdős #728: first autonomous solve, with a Lean certificate

T2

Feb 2026

Gauge-theory amplitudes: GPT-5.2 and five named physicists

PHYSICS

T2

May 2026

Unit distance disproved: nine mathematicians read it line by line

T2+

Jun 2026

Jamming exponents, a 12-year-old identity: Parisi (Nobel) and Zamponi with Claude, over forty rounds of dialogue. Peer reviewed, *J. Stat. Mech.*

PHYSICS

T2

Jul 2026

Jacobian conjecture false in $\dim \geq 3$: announced in one post on X

T3

Jul 2026

Complex structure on S^6 , open since 1947: 108 pages, self-hosted, no arXiv

T2

Jul 2026

HAWK-256 broken: a CWI lattice signature withdrawn from NIST within a day
Mythos Preview found a nontrivial lattice automorphism, 2^{64} to 2^{38} . The authors confirmed and withdrew, 29 Jul

CRYPTO

T1

Aug 2026

Two thirds of the zeta zeros on the critical line, up from 41.6%
arXiv comments: "Proof discovered autonomously by Claude". Alpöge and Furman; Lean; read by Conrey and Goldston

T1*

Aug 2026

"Astra": ten decade-open problems, Lean, <\$2k each (zero referees)

T2

Aug 2026

Entanglement of formation: Holevo and Shirokov find a counterexample
"...the help of Claude Fable 5 to find a state ρ ": from the abstract, quant-ph, 27 Aug

QUANTUM

...and the denominator

The community's own tracker flagged the AI contributions nobody had checked, then stopped updating on 30 June.

"AI is being used a lot by people who aren't mathematicians... no human has read it."

Thomas Bloom, curator, erdosproblems.com

When it froze, 20 of its 61 "AI standalone" entries were marked incorrect or unverified.

1 in 277

1 in 2,828 (2023) to 1 in 458 (2025) to 1 in 277 (2026): arXiv submissions with fabricated citations. Now a one-year ban.

"hello there the jacobian conjecture is false thanx"
the entire public announcement, 20 Jul 2026. An 87-year-old conjecture, checked by hand within days
The same author's 108-page S^6 paper: self-hosted, no arXiv, and reportedly written out by Claude

Generation has been automated faster than verification.

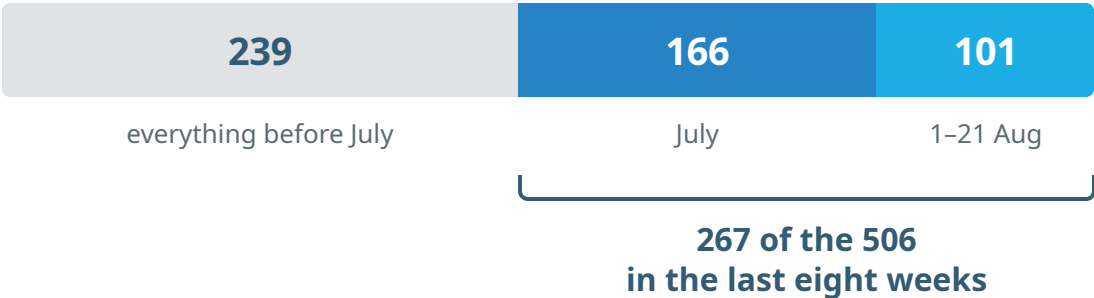
Now somebody has to count it

ten independent registries,
none older than eight months · 2 Sep 2026

HOW FAST

506 claimed AI-solved results

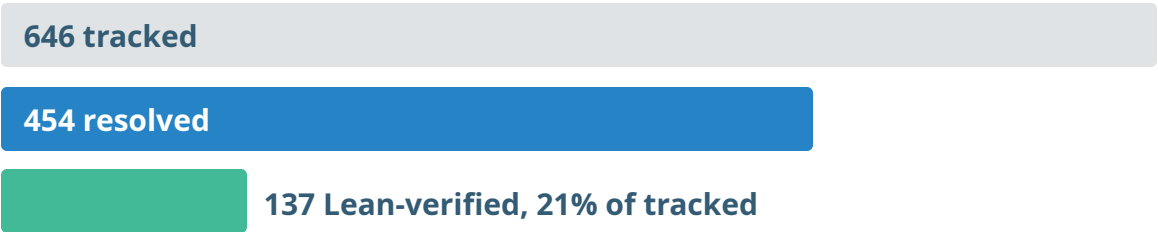
aimath.robertj1.com, one registry, one field. Snapshot of 21 Aug 2026.
Still 506: this registry has not been updated since 21 August.



HOW MUCH OF IT IS CHECKED

454 resolved, 137 machine-checked

vibemathed.com, a second registry, tracking 646 problems. As of 2 Sep 2026.



17 new entries in the last week, down 43% on the week before. Two weeks of decline.

One wiki page could not keep up. Five more of them, all built in the last few months, each counting something different.

mathdb.com

87,014

open problems catalogued, with no
verification tags anywhere

theoremdb.org

2,322

problems reviewed, still 31 solved.
MCP endpoints so agents can file
results

Epoch AI

5 of 50

curated research problems solved by
AI (1 by a human, 44 open). The
strictest denominator anywhere

Palomar

34

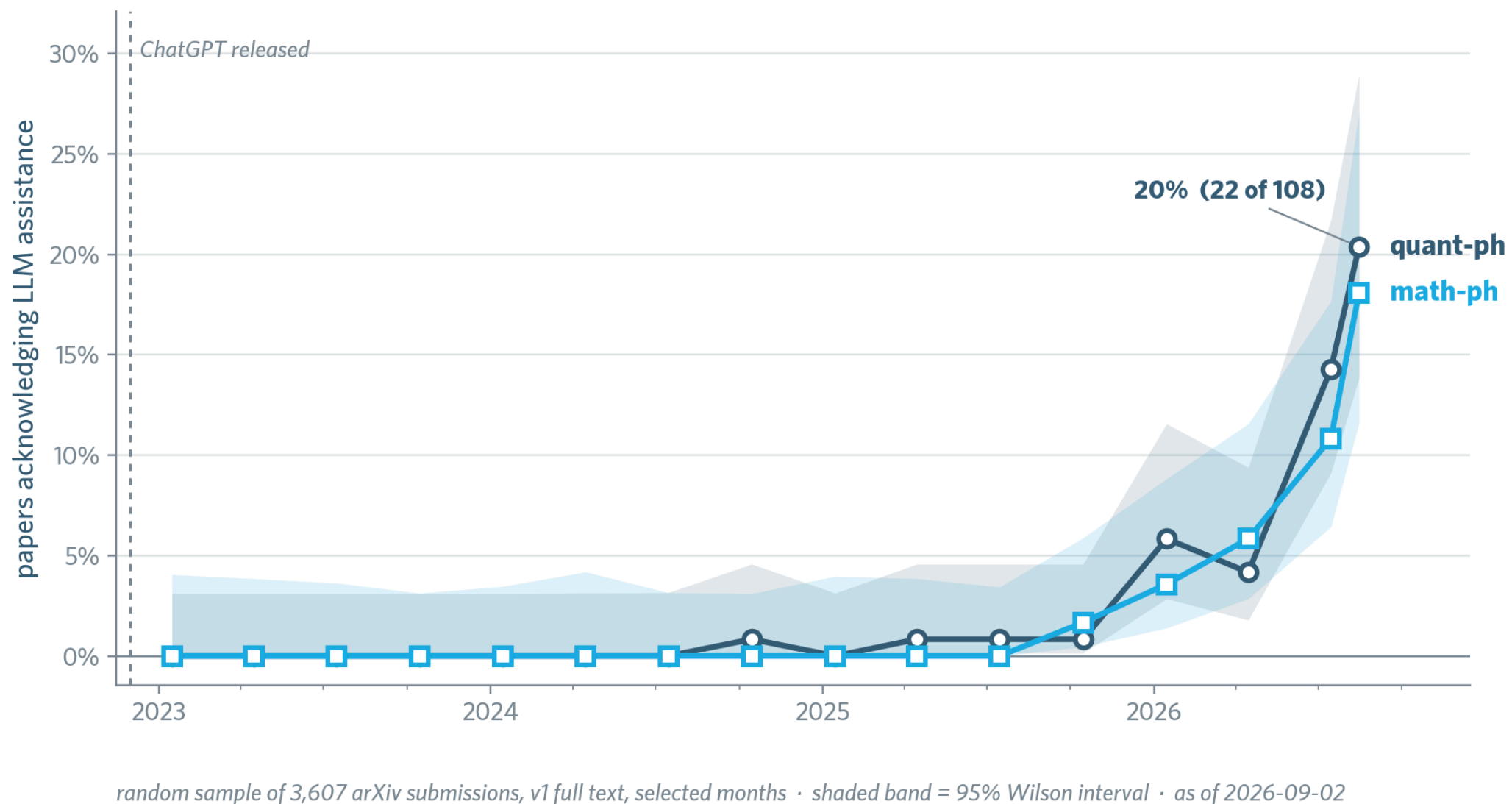
entries since 18 Aug, thirty of them in
three days. Tao's Lean-proof registry;
an LLM checks the formal statement
against the informal claim

openproblem.ai

10

published claims, 0 verified. The
strictest registry is the smallest

Ten ways to count the flood. Not one of them is peer review.



Papers whose acknowledgments name an LLM: quant-ph **0% through 2023, 20% in August 2026**
3,607 sampled v1 submissions, every hit read by hand · a disclosure rate, not a usage rate · 2 Sep 2026

A control experiment, run on us

First-year theoretical physics, six years of grades

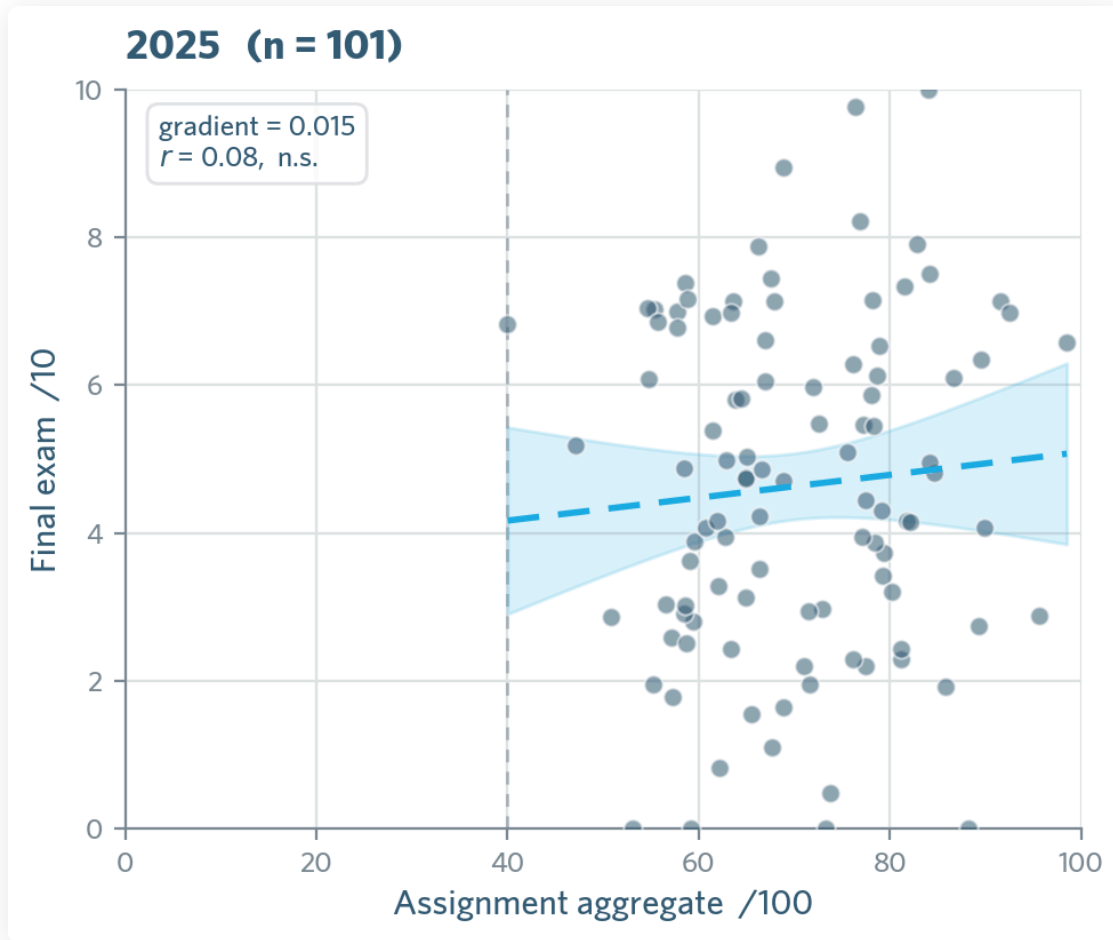
Weekly take-home assignment, out of 100 (unsupervised)

Invigilated final exam, out of 10 (the control)

2020 to 2023: the two agree at **$b \approx 0.10$**

Synthetic data, illustrative: reproduces the structure of the real six-year analysis.

Six years, one control experiment



2025

GRADIENT B

0.015

CORRELATION

R

0.08

$r = 0.08$, not significant: the assignment no longer measures what the exam measures.



press → or drag to step through the years

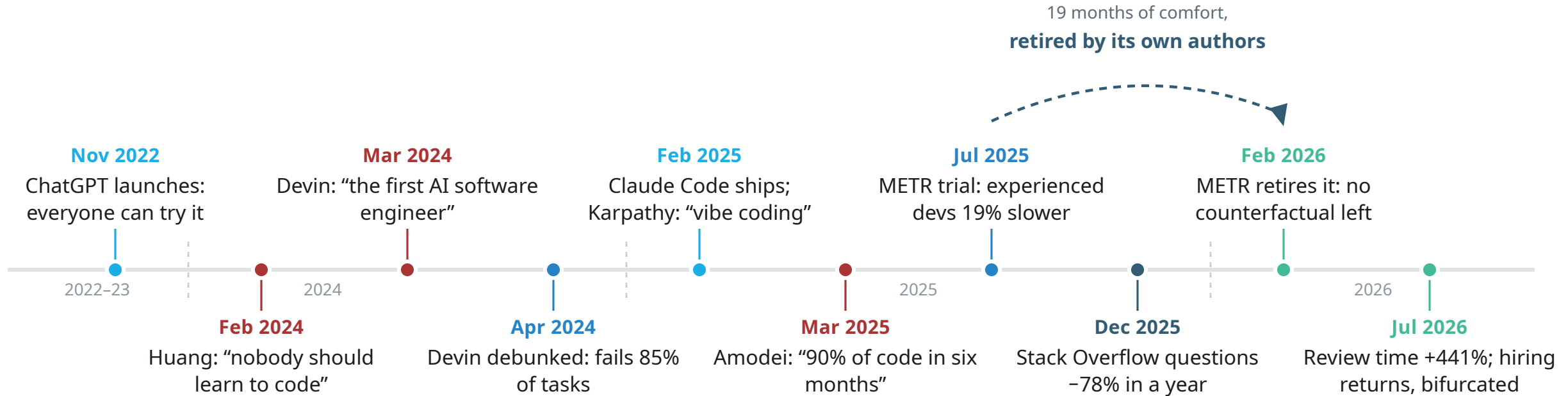
Synthetic data, illustrative: reproduces the structure of the real six-year analysis.

Quick survey: hands up

- Who has used ChatGPT, Claude, or Gemini?
 - Who has used one for research?
 - Who has used a coding agent?

The timeline

software engineering
Nov 2022 → Jul 2026



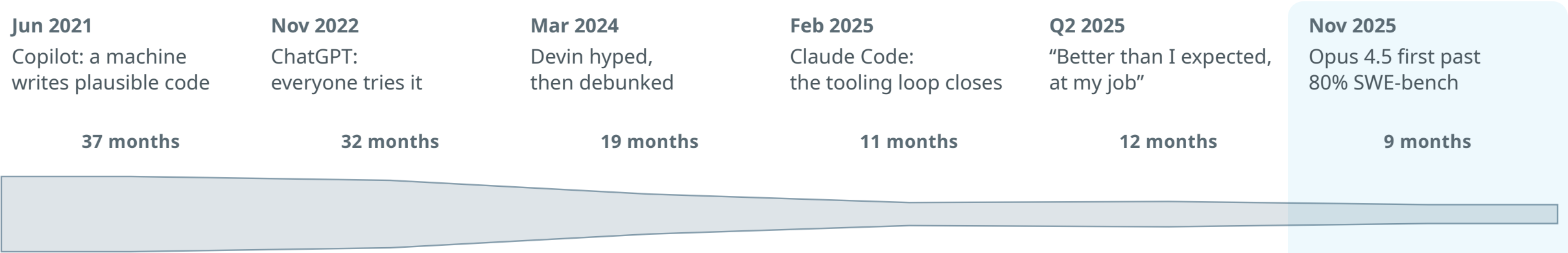
Software engineering has already run this whole arc

Four years, launch to routine use, with a very loud detour on the way.

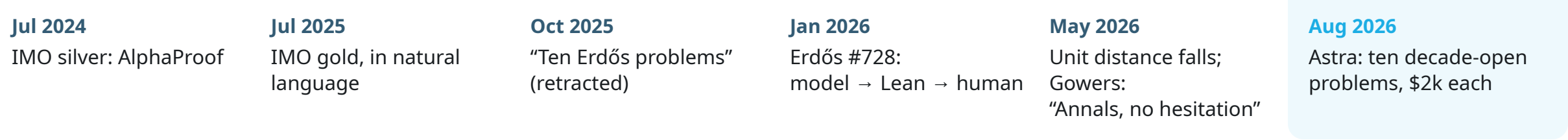
Theoretical sciences are 18 months behind

the same arc, one field later
2021 → 2026

SOFTWARE ENGINEERING



MATHEMATICS & THEORETICAL PHYSICS



You are here

4 September 2026, this room

Offset ~3 years in 2024 · ~9 months now · closing at about a year per year

Progress is astonishing.
However, the failure modes have not gone away.

To understand them, we have to look at the architecture.

The gap between

"I tried ChatGPT and it hallucinated"

and

"this is a powerful force multiplier"

is not closed by better models.

It is closed by **technique** and **domain expertise**.

PART TWO

What is an LLM?

**A new kind of resource:
general-purpose intelligence,
bought by the token**

(kind of)

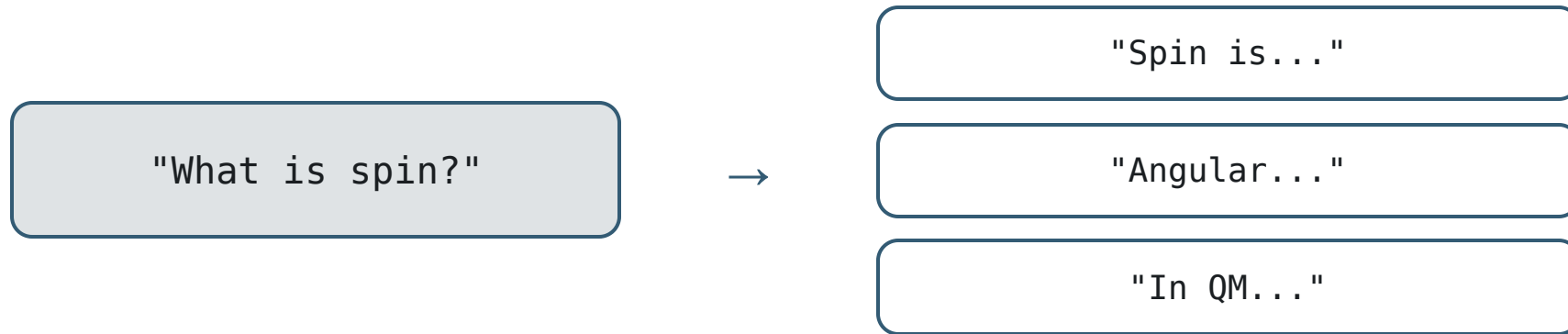
From outside, looking in

```
f : String -> String
```

Outside the API boundary, this is the observable behaviour.

No hidden state is carried between calls.

Nondeterministic



Same input, different outputs: the model samples from **$P(\text{next token} \mid \text{context})$**

Temperature

$$P(\text{token}_{\textcolor{red}{i}}) = \exp(\ell_{\textcolor{red}{i}}/T) / \sum_{\textcolor{red}{j}} \exp(\ell_{\textcolor{red}{j}}/T)$$

spin		0%
charge		0%
mass		0%
momentum		0%
energy		0%
a		0%
color		0%
flavour		0%

Temperature 1.0



Sample

Autoplay

The electron has

Direct analogy to the Boltzmann distribution. Low T freezes into the ground state; high T melts toward uniform.

Tokens

Not characters. Not words.

Subword chunks from a fixed vocabulary **V** of about 100k entries.

```
'theoretical physics' -> ['theor', 'etical', ' physics']
```

More precisely: **f** : **Token*** → **Token***

An LLM is a stateless, nondeterministic function

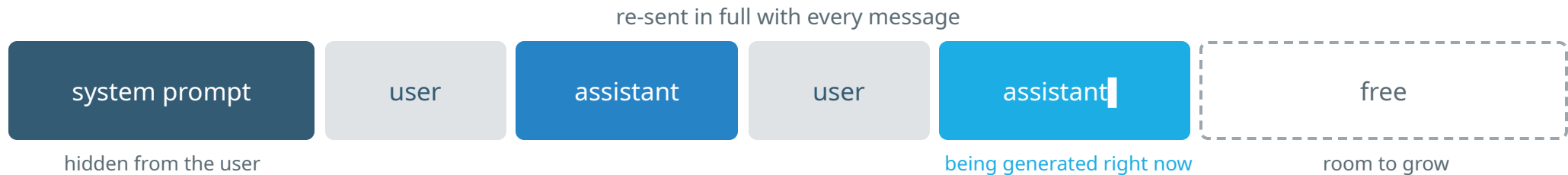
f : String -> String

that samples from $P(\text{next_token} \mid \text{context})$

via a single forward pass through a frozen neural network

The context window

Everything the model sees is **one token sequence**, capped at a fixed length: the **context window**



Current models: **200k to 1M tokens** (a million tokens is roughly two thousand pages)

Whoever controls the context controls what the model knows

PART THREE

The illusion of chat

A single API call

```
curl https://api.anthropic.com/v1/messages \
  -H "x-api-key: $KEY" \
  -H "content-type: application/json" \
  -H "anthropic-version: 2023-06-01" \
  -d '{"model": "claude-sonnet-5", "max_tokens": 256,
      "messages": [{"role": "user", "content": "What is a quark?"}]}'
```

One string in, one string out: the entire interaction

How chat works

Turn 1: [user_1]

Turn 2: [user_1, asst_1, user_2]

Turn 3: [user_1, asst_1, user_2, asst_2, user_3]

The client re-sends the **entire** history every turn: the LLM has no memory.

INTERLUDE

Failure modes

3½

The failure zoo: architectural

Hallucination

No way to guarantee the fidelity of information stored in the weights, and what comes out is *sampled*, not looked up.

Training cutoff

The weights are frozen once training ends. No continual learning: recall of anything later is guesswork.

Nonzero error rate

Every token is drawn from a probability distribution. Prose tolerates that; deduction and code do not.

Context rot

Attention degrades as the window fills. In our experience quality starts to sag beyond roughly **30%** usage.

Four consequences of the architecture, not four bugs.

The failure zoo: post-training

Hyperfixation (reward hacking)

Rewarded for visible completion: it deletes the failing test, hard-codes the answer, declares success.

Sycophancy

Raters prefer being agreed with. So it preferentially asserts the falsehoods *you would quite like to be true*.

Confident wrongness

Fluent, specific, real-shaped, false, and unhedged.
Never ask a model to check its own output.

Semantic shift

It imports assumptions from a better-studied neighbour: Jordan algebras silently treated as C*-algebras.

These four are trained in, and they compound with each other.

These are **architectural facts**, **not mysteries**

A frozen network generating tokens by sampling,
steered by whatever happens to occupy its context.

And each one admits a structural countermeasure.

There is a way to correct for these errors

Reliable computation out of unreliable components.

PART FOUR

From function to agent

The agent loop

```
while not done:

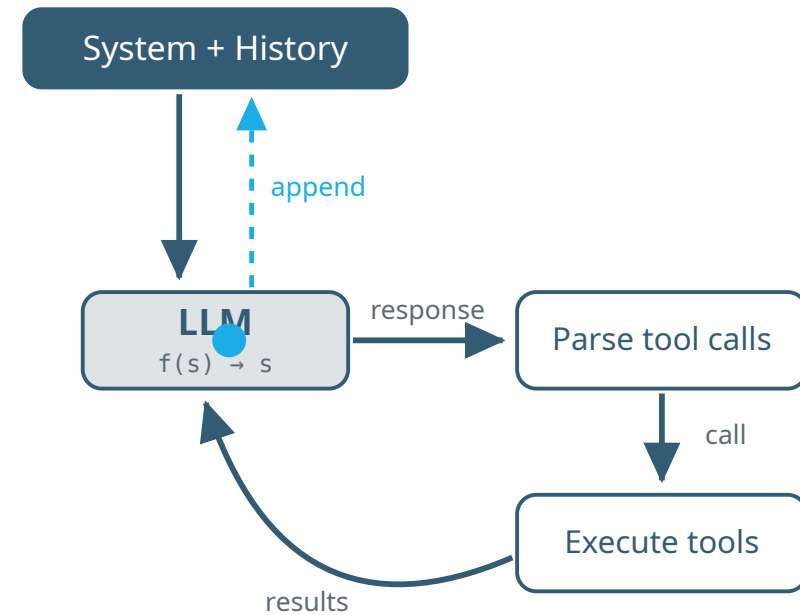
    response = llm(system_prompt + history)

    tool_calls = parse(response)

    results = execute(tool_calls)

    history += [response, results]
```

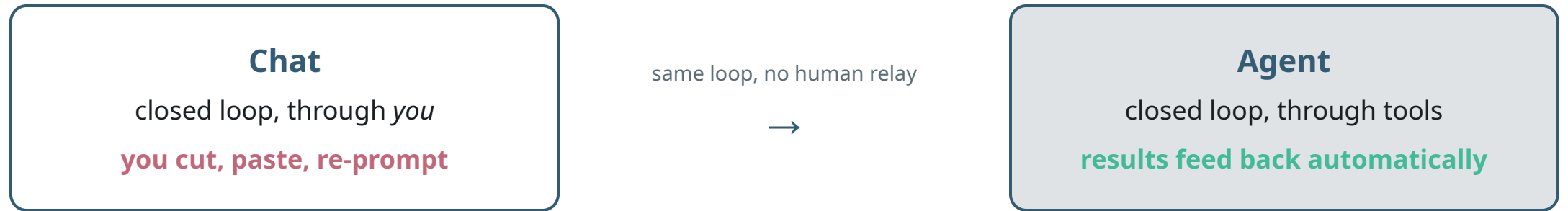
The complete architecture of every coding agent



The LLM never executes anything

It generates text; your scaffolding acts on it.

Closing the loop



Chat is a feedback loop too: you are the wire. A failing test, a stack trace, a wrong number: each flows straight back into the context.

The agent closes the same loop faster, with no human cutting and pasting

Error correction for LLMs

FAILURE MODE	COUNTERMEASURE
Finite context · no continual learning	→ filesystem as memory
Hallucination	→ ground truth on disk (every cited paper kept locally)
Nonzero error rates	→ recursive decomposition and verification
Hyperfixation · sycophancy	→ independent adversarial verification (in a fresh session)
Confident wrongness	→ structured proofs (Lamport)
Semantic shift	→ formalisation
Lack of taste	→ ask a human

Every cure on the right is a tool the agent loop already gives us

The filesystem is permanent memory

Chat

context is the only state

it evaporates when the window fills

rots, cannot be checked

Agent

the filesystem is the state

it survives the context window

persistent, checkable, versioned

(i) memory that outlives the context window; (ii) ground truth the model can **re-read** instead of half-remembering.

The radical unlock versus chat

Errors compound

A flat chain of thought has **no error detection**. If each step succeeds with probability $1 - \epsilon$:

$$P(\text{the whole derivation}) = (1 - \epsilon)^n$$

PER-STEP ERROR

1%

STEPS

100

SURVIVE INTACT

37%

A slip at step 3 propagates silently to step 40, while the trace stays locally fluent throughout.

This is why long unchecked derivations rot

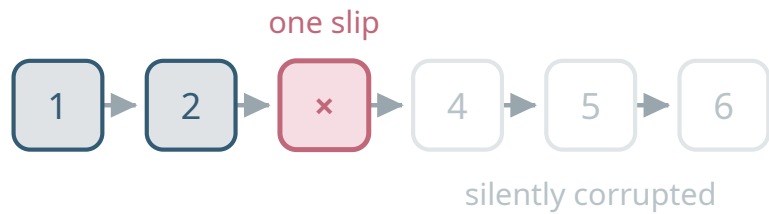
Subagents



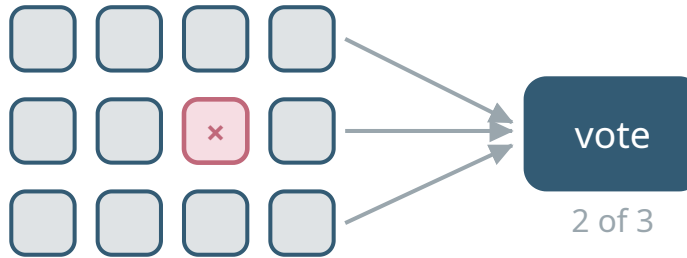
Each worker gets a clean window; results return as short reports.

Context isolation is the point

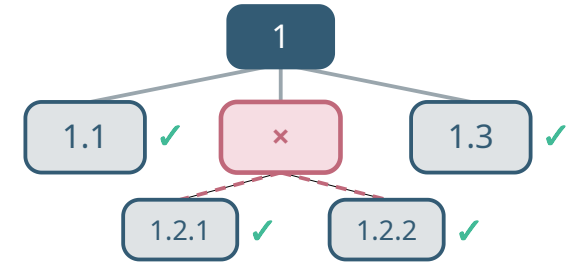
Three ways to spend compute on reliability



(a) flat chain. One slip at step 3; everything after it is quietly wrong.



(b) parallel repetition. Majority vote over final answers. Helps, but scales poorly with depth.



(c) recursive decomposition. Verify each block in isolation; expand the one that fails.

(c) is the concatenated-code strategy (von Neumann, 1956).

Above the verification threshold, reliability *compounds* instead of decaying

Why structure a proof?

Most proofs in physics and mathematics are written as flowing prose:

“By linearity ... and therefore ... which contradicts ... hence the result follows.”

This works well enough when the proof is short. But even a five-line argument can hide a gap, one that only surfaces when someone pushes back on a specific step.

And then the reader is left asking: *which* step was wrong?

Lamport structured proofs

```
1      the claim
      1.1      ASSUME ...
      1.2      ...                                [1.1]
            1.2.1      ...
      1.3      QED                                [1.2]
```

Every assertion gets a **hierarchical number**: 1, 1.1, 1.1.1 ...

Every step is justified: by citing earlier steps, or by its own children.

ASSUME / **CASE** / **QED** open and close scopes.

A leaf with no justification is an **exposed gap**.

Auditable

"I don't believe 1.3.2"

Modular

sub-steps refine independently

Gap-visible

nowhere for a hole to hide

That is the entire formalism: no logic syntax to learn

*L. Lamport, "How to Write a 21st Century Proof", J. Fixed Point Theory Appl. **11**, 43–63 (2012)*

No-cloning, structured

1 **There is no unitary U with $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ for every state $|\psi\rangle$.**

1.1 **ASSUME** such a cloning unitary U exists.

1.2 Basis states clone: $U|0\rangle|0\rangle = |00\rangle$ and $U|1\rangle|0\rangle = |11\rangle$. [1.1]

1.3 By linearity: $U|+\rangle|0\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. [1.2]

1.4 By cloning: $U|+\rangle|0\rangle = |+\rangle|+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$. [1.1]

1.5 Contradiction: Schmidt rank $2 \neq 1$, entangled on the left, product on the right. [1.3, 1.4]

1.6 **QED** The assumption 1.1 is false: no such U exists. [1.5]

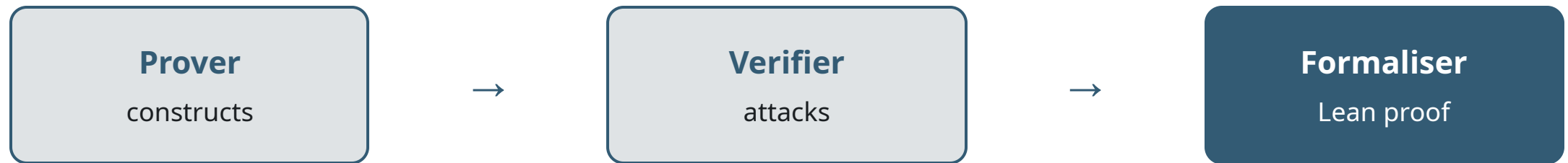
Doubt step 1.5? Expand it: the proof refines exactly where the sceptic pushes.

1.5.1 Arrange the four coefficients as a 2×2 matrix.

1.5.2 Its rank is 2 on the left, 1 on the right.

Each numbered step is the small, independently checkable block the error-correction recipe demands.

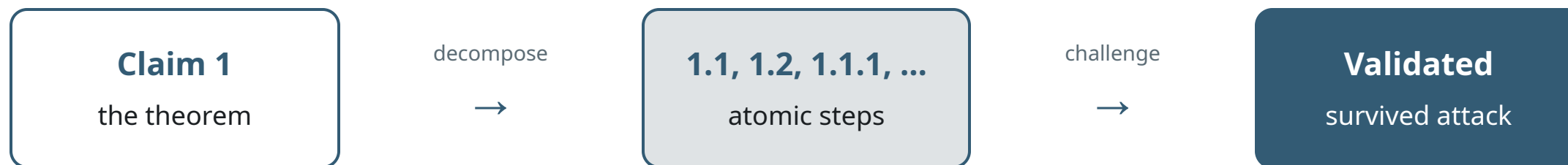
Adversarial verification



Natural language is an inadequate verification mechanism

Formal proof = a compiler for mathematics

Vibefeld: molecular pedantry



A natural-language proof, decomposed into a ledger of tiny numbered claims; each must survive adversarial challenge *on its own*.

A hallucinated proof has nowhere to hide

github.com/tobiasosborne/vibefeld

Break the correlations

A verifier only helps if its blind spots are not the prover's blind spots. Three knobs:

Blind the verifier

It sees the numbered artifact *only*, never the prover's reasoning.

Change model family

Claude checks GPT. Different labs, different training data, different failure modes.

Change capability class

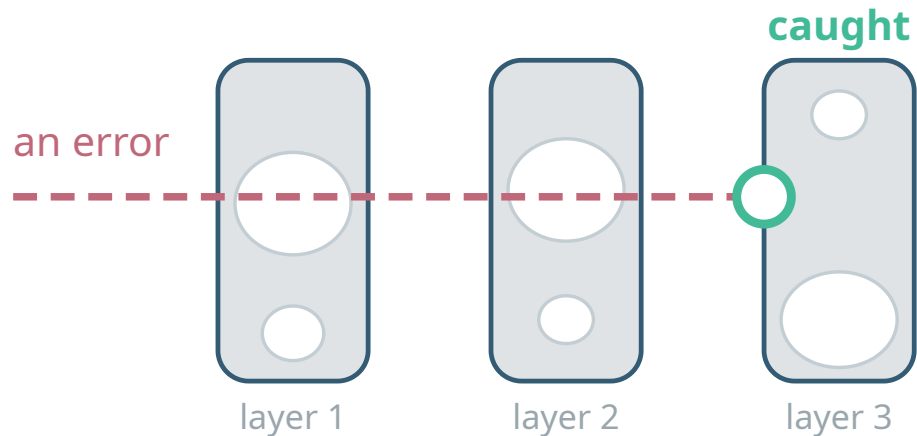
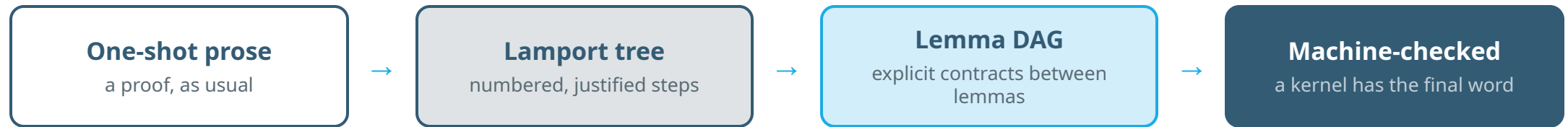
A frontier model audits the output of a cheaper, faster one.

A blind spot shared by every fresh session of one model is often glaring to another

Agent harnesses are ordinary command-line programs: one agent invokes another through its shell tool, so this automates.

Lean can be reward hacked

Formalisation is not a single act. It is a spectrum of artifacts, increasing in cost and in rigour:



No single layer is trustworthy alone. But the layers fail in **different places**: compose them. Swiss cheese (Reason, 2000).

An agent produces every layer almost for free

What should a referee now demand?

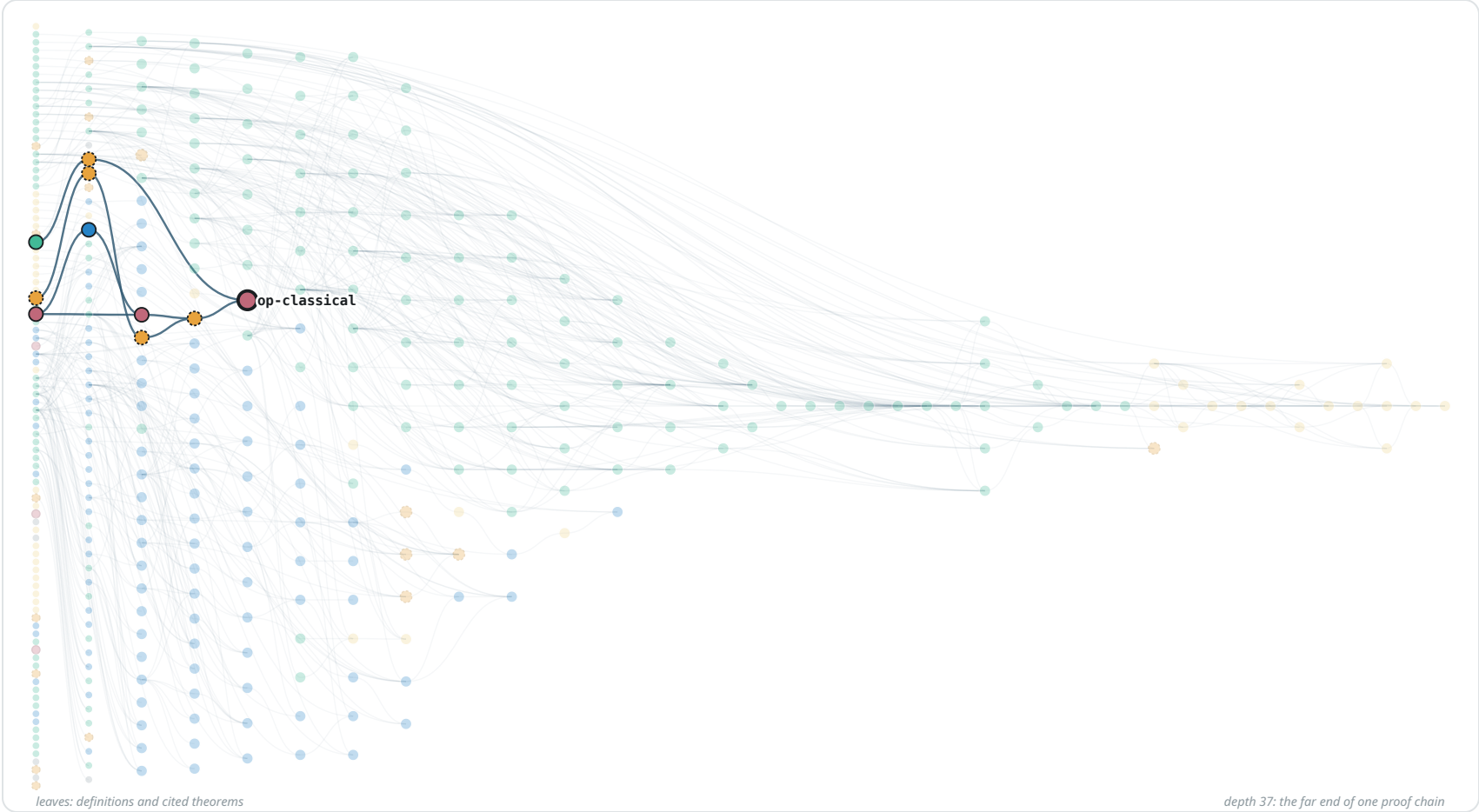
WHAT IS CLAIMED		WHAT SHOULD ARRIVE WITH IT
A theorem	→	a numbered proof tree (gaps visible, not prose)
A hard lemma	→	a machine-checked certificate (or a stated reason there is none)
A counterexample	→	the witness, runnable (code, seed, data)
Any of it, produced with an agent	→	the disclosure, and which steps it wrote

None of this is new. It is what a good referee always wanted and could not previously ask for at a reasonable price.

Raise the evidentiary standard, because producing the evidence just got cheap

What the Swiss cheese produces

one open problem, worked by agents for five weeks
364 results · 769 dependency edges · every arrow is a checked contract



click any node: what it rests on lights up · depth grows to the right

● machine-validated: proved by one fresh agent, attacked by another	169
● proved in prose, not yet adversarially checked	113
● inherited paper-proof, unaudited here	22
● conjecture, stated, heuristic or numerical	48
● open problem	6
● disproved or obstruction	6

op-classical

OPEN-PROBLEM

OPEN

(OPEN) Classical projection stability: there are universal $\eta_0, C > 0$ (n-free) such that every row-stochastic Q with $\|Q^2 - Q\|_{\infty} \leq \eta \leq \eta_0$ admits a stochast...

rests on 9 results: 1 machine-validated · 1 prose · 7 unverified or open

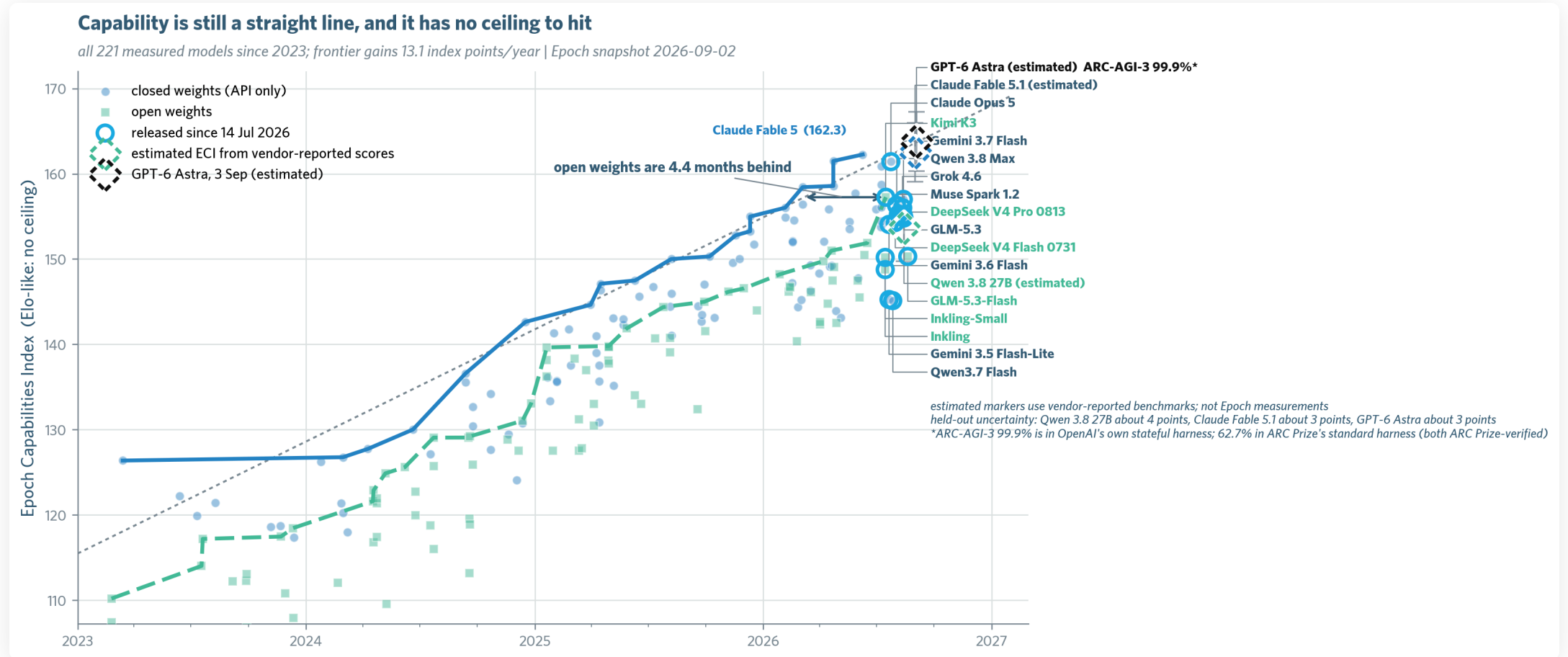
The open problem itself. What it still rests on is orange and red: the gap is visible, named, and exactly where the next agent gets sent.

One agent proves, a fresh one attacks, a linker keeps the ledger. The colour is what survived.

PART FIVE

The trend

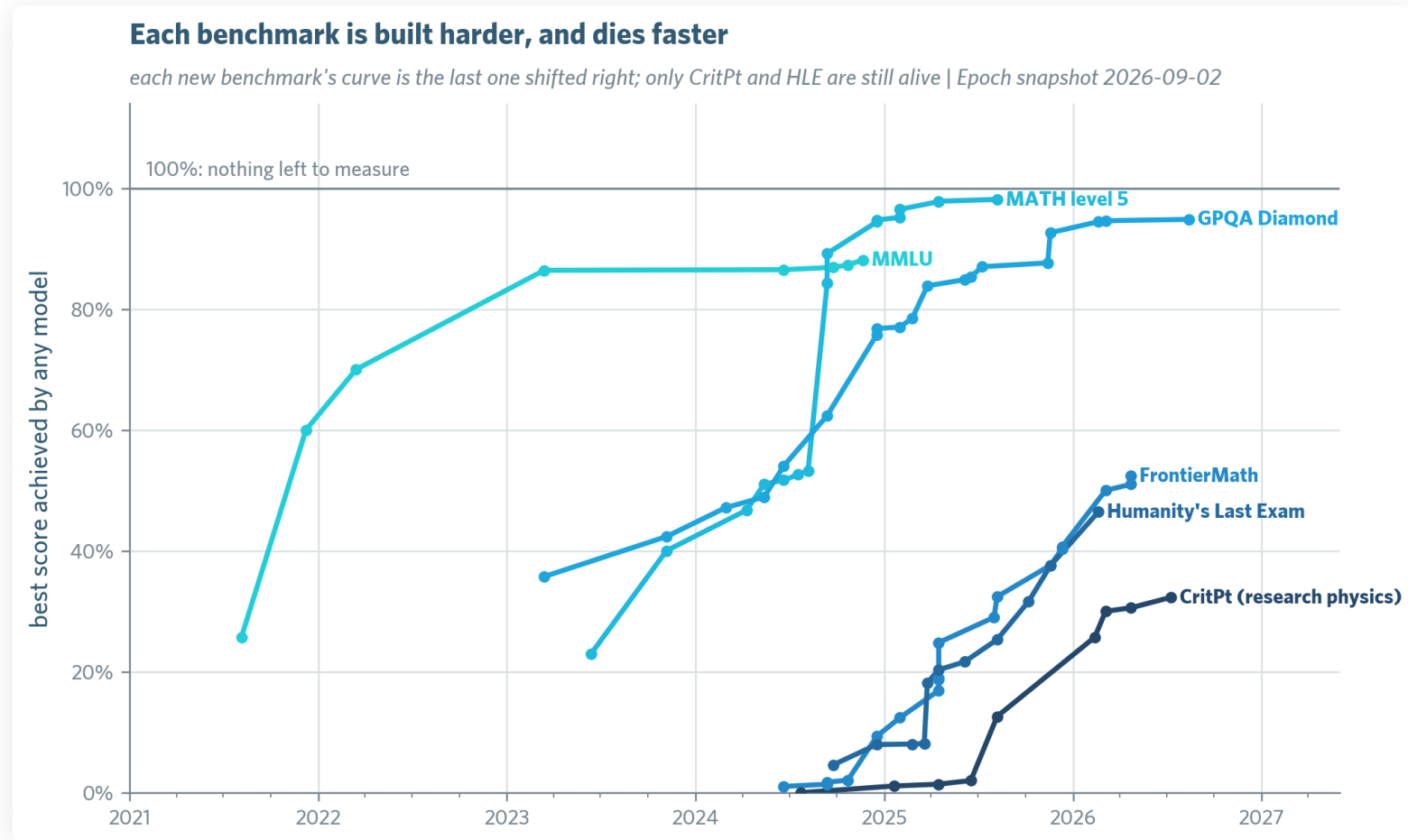
Capability is a straight line



13.1 ECI points per year; open weights about 4.4 months behind (Kimi K3). **GPT-6 Astra (black, released last night)** saturates ARC-AGI-3 in OpenAI's own harness, 62.7% in ARC Prize's; its position and Fable 5.1's are estimates from reported scores, not Epoch measurements.

Epoch AI Benchmarking Hub, CC-BY, snapshot 2026-09-02

Benchmarks saturate, then get replaced



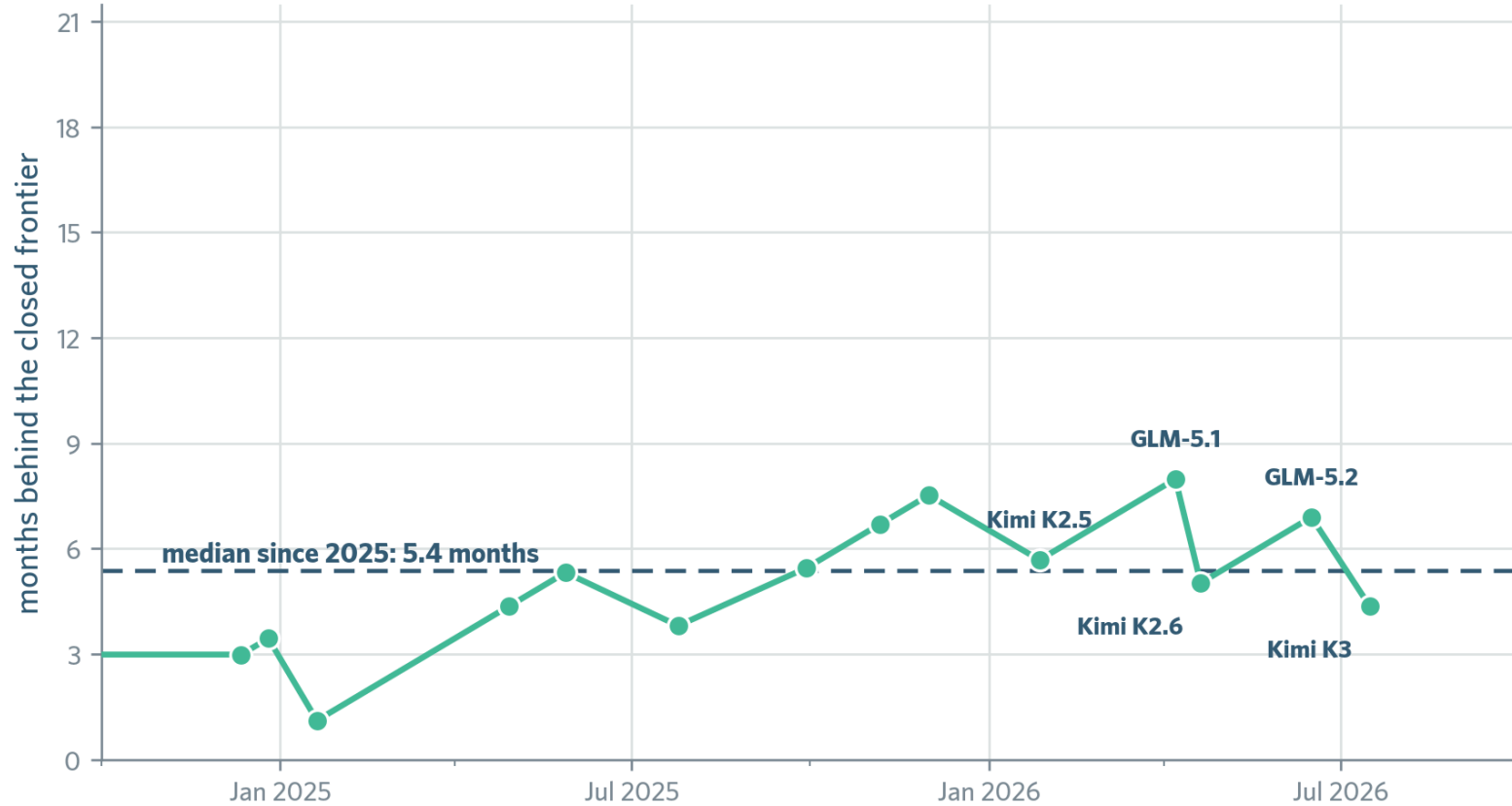
GPQA (PhD science QA) is **dead at 94.8%**; research physics (CritPt) still at **32%**; FrontierMath's original tiers are frozen at 52% because Epoch replaced them with harder v2 tiers

Epoch AI Benchmarking Hub, CC-BY, snapshot 2026-09-02

You can run the frontier locally

Open weights trail the frontier by about 5 months

how long the closed frontier had already been at the level each new open model reaches | Epoch snapshot 2026-09-02



Qwen 3.8 27B: the first truly capable model that **fits one consumer GPU when quantised** (RTX 3090, 24 GB)
no vendor lock-in · Qwen ECI estimated from vendor-reported scores · Epoch AI Benchmarking Hub, CC-BY, snapshot 2026-09-02

The open frontier is already hosted in Europe

Model	Made by	Hosted by (EU)	€ per M tokens in / out
Kimi K3	Moonshot (CN)	Cortecs, "Sovereign Cloud"	2.69 / 13.46
DeepSeek V4 Flash	DeepSeek (CN)	Cortecs	0.12 / 0.25
GLM 5.3	Z.ai (CN)	Cortecs	1.26 / 3.95
Qwen 3.8 27B	Alibaba (CN), Apache 2.0	Cortecs, EUrouter (Amsterdam)	0.30 / 2.20

Claude Fable 5.1 on Anthropic's API: \$10 / \$50. The best open model on EU hardware costs about a quarter of that; the cheap one, about 1/200.

Cortecs (Vienna): OpenAI- and Anthropic-compatible endpoint, GPUs in ES, DE, FR, FI, PL. Also: EUrouter (Amsterdam), Mistral La Plateforme, Scaleway, OVHcloud, IONOS, STACKIT. Coming: the Dutch AI Factory in Groningen, about 1,800 GPUs, 2027.

cortecs.ai/serverlessModels · eurouter.ai · mistral.ai · epoch.ai/data-insights/open-closed-eci-gap · surf.nl

Who owns the instrument?

Rent the frontier

Best capability, today, by the token. Three US vendors. Terms that can change in a week, and every prompt leaves the building.

Host open weights

Kimi K3, DeepSeek, GLM, Qwen: about 4.4 months behind the closed frontier, on hardware in the EU, or SURF's. Read the licences. And note who made them: the open frontier is made in China; Europe's own frontier-scale open model is Mistral Large 3.

Run it on your desk

Qwen 3.8 27B on a single 24 GB card. Weaker, private, and yours. The fallback that cannot be revoked.

Sovereignty here means jurisdiction and data residency, not provenance. The price of it is about a third of a year of capability, and it has been shrinking.

Choose the column deliberately, before a change of terms chooses it for you

What it costs, on your side of the ledger

\$20 / month

chat only
where most of your people still are

\$100-200 / month

a working agent seat
the tier everything in this talk needs

< \$2,000

one decade-open conjecture
Astra, August 2026

~\$100,000

one NIST candidate broken
HAWK-256, 60 hours of agents

Forty agent seats for a year is about **\$96,000**: roughly one postdoc. The instrument currently costs less than the person driving it.

Inference is a consumable, not capital. It behaves like liquid helium, not like a cryostat, and no grant template has a box for it.

Budget it per project, per year, and assume the unit price keeps falling

Every layer we built today has failure modes

Context rot, hallucination, tool misuse, prompt fragility

The human in the loop is not optional

**You can outsource cognition
but you can't outsource *understanding***

Summary

LLMs are stateless functions: not magic, not minds

Agents are loops: scaffolding provides the action

The filesystem is ground truth: memory that survives and can be checked

The human provides taste: technique over hype

Try it this week

Claude Code

terminal agent
anthropic

Codex

terminal agent
openai

Kimi K3 via Cortecs

open weights, hosted in the EU
cortecs.ai

Bring a real research problem, not a toy

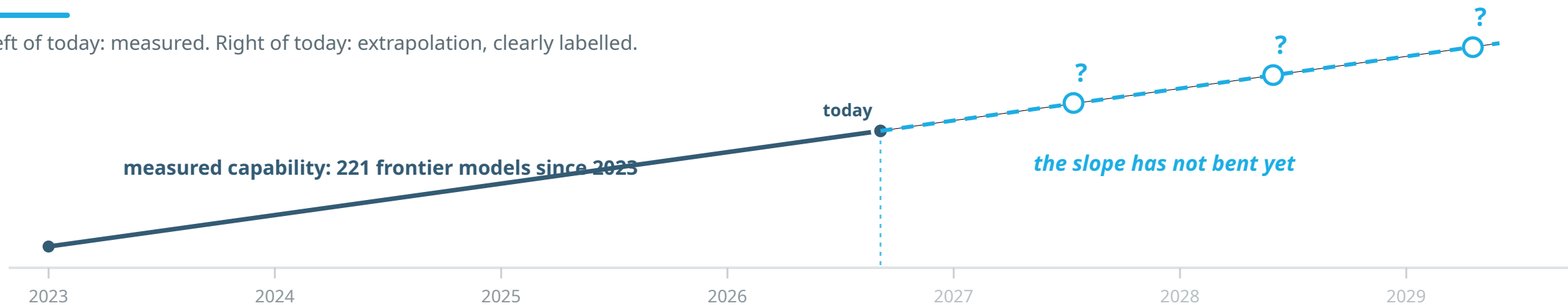
An afternoon suffices

- 1 Open a coding agent in a directory you care about.
- 2 Write a **CLAUDE.md** recording your standards: rigour, spelling, what counts as a source.
- 3 Keep a worklog file, and a directory holding every paper you cite.
- 4 Take the lemma you are currently fighting and write it as a Lamport tree.
- 5 Set one agent to prove and a second to attack, and only then read the result.
- 6 Formalise the one step you trust least.

The first three need no programming at all

The timeline, extended

Left of today: measured. Right of today: extrapolation, clearly labelled.



+13.1 / yr

Epoch Capabilities Index:
still linear, no ceiling in sight

94.8%

GPQA Diamond, saturated:
the expert humans sit at 69.7%

32.3%

CritPt research physics:
your frontier is still wide open

~1000×

Cheaper per attempt than 2023,
so run it a hundred ways and check

\$2,000

Price of a decade-open
conjecture, August 2026

2 s → 14 h

Autonomous task horizon (METR):
doubling every 4–7 months

Who did well in software: not the people who denied it, not the people who surrendered to it.
It was the people who learned to specify, to check, and to own the result.

What would you attempt if the tedious 90% were free?

Aim there.

The nostalgic version

Stacey Jeffery · CWI and QuSoft, University of Amsterdam · LinkedIn, 26 Aug 2026

“Yesterday I put a paper on the arxiv, and it might be the last paper I ever write in which all ideas are human. Feeling a bit nostalgic.”

The paper, with Freek Witteveen: a new quantum algorithm for ground-state energy estimation, matching a lower bound posted the same day by Somma and de Wolf. arXiv:2608.24494, 25 August.

The blunt version

Alex Kontorovich @AlexKontorovich · X, 26 Aug 2026

“So... figure out how you can be useful to others! Nobody said it was easy. You’re not ‘owed’ a job by society, no matter how many years of training/experience you’ve invested.”

“We are granted tenure not so that we can sit on our laurels, but so we can take really big risks that may not pan out for a long time (if ever).”

The gentler version: stay curious, and go looking for the hard problems

Finding them means knowing what these systems can and cannot do. That is what today was for.

**The slop cannon is charged
whether we like it or not.
The only question is **how it is aimed****

Aimed with discipline (error-corrected, ground-truthed, adversarially checked),
it is the cheapest instrument for rigorous exploration the sciences have ever been handed.

Raise High Your Slop Cannon (guide in preparation, 2026)

ADDITIONAL SLIDES

But don't they lose money?

B

Triangulating the true cost of inference

Many independent, differently-biased estimators, combined in log-space: energy physics (lower bound) · GPU rental + throughput · open-weight market prices (upper bound) · DeepSeek's production disclosure · first-principles models (Epoch AI) · leaked P&Ls

\$0.35 / M

DeepSeek-V3 class
37B active, MLA
68% CI \$0.27–0.46

\$2.1 / M

frontier workhorse
GPT-5 / Sonnet class
68% CI \$1.2–3.7

\$8.5 / M

frontier flagship
Opus class
68% CI \$3.5–21

A typical chat query costs the provider ~0.1 cents

folklore is off ~10x on energy (0.3 Wh measured vs. the “3 Wh” meme), ~100x on cost

Prices vs. costs: the margins are real

86%

Claude Sonnet class

\$15 / M output

[57–95%], $P(>0) > 99.9\%$

79%

GPT-5 class

\$10 / M output

[35–93%], $P(>0) = 99.7\%$

84%

DeepSeek R1, \$2.19 / M

self-disclosed

“545% cost-profit ratio”

Leaked P&Ls agree: Anthropic gross margin –94% (2024) → mid-60s% (Q1 2026), first operating profit Q2 2026; OpenAI 28% → 43% while free users consume ~half the inference bill

“We're profitable on inference.” (Altman, Aug 2025) “>50%” (Amodei, Feb 2026)

The honest fine print

Subscriptions: the median \$20 subscriber costs *cents* per month; <5% hit rate caps

\$100–200 power tiers are bimodal (Altman, Jan 2025: Pro lost money), and rate limits cap the tail

What survives on the skeptic side:

long-context cost floors · reasoning models burn 10–50x tokens per query · GPU rents *rose* in 2026

The losses are real, but discretionary:

OpenAI 2025 operating loss ~\$21B = R&D \$19B + training ~\$8B, on *positive* serving margins

Not failing to make money: choosing to reinvest more than they earn